Ayushmaan Puri

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Education

University of California, San Diego — Expected August 2026
B.S. CSE: Computer Engineering — *GPA 3.50/4.0*
Relevant Coursework: Computer Architecture, Parallel Programming with OpenCL, Embedded Systems Design, Operating Systems, Digital Design with SystemVerilog, Digital Circuits, Signal Analysis, Advanced Data Structures and Algorithms

Research Experience

Undergraduate Researcher | Junkyard Autograder — April 2026 – Present
Kastner Research Group – UC San Diego — *La Jolla, CA*

- Investigating repurposing discarded smartphones as a dedicated autograding cluster to replace oversubscribed GPU pods shared across a class of 330 students.
- Deployed a 40-node k3s Kubernetes cluster on ARM64 Google Pixel Fold devices running Debian over USB Ethernet; configured each node as an isolated Linux execution environment for containerized grading jobs.
- Modeled real submission timing and runtime distributions from 8 programming assignments using Gaussian Mixture Models with BIC-selected component counts.
- Built a synthetic workload generator sampling from fitted GMMs to stress test cluster throughput and determine the minimum phone count needed to serve a class without queuing delays.

Undergraduate Researcher | UnifiedSplitting — June 2025 – March 2026
PatLab & UC Scholars Research Program – UC San Diego — *La Jolla, CA*

- Built heterogeneous CPU-GPU pipelines for efficient CNN inference on unified memory architectures.
- Implemented model-splitting strategies for parallel CPU-GPU inference on NVIDIA Jetson devices.
- Profiled per-layer CPU-to-GPU speedup ratios across full network architectures, identifying a 10x gap between convolutional layers (300–500x) and fully connected layers (30–40x) that informed the optimal splitting point.
- Validated the approach across five architectures (ResNet-18/50, MobileNetV2, EfficientNet-B2, ViT-Tiny) and six simulated edge hardware profiles, achieving 8–24% throughput gains over GPU-only baselines.
- Wrote and presented a methods paper at UCSD Summer Research Conference 2025 and Southern California Conference for Undergraduate Research 2025.

Summer Research Intern | Active Thermography for Defect Detection — June 2024 – Sep. 2024
SeNSE – Indian Institute of Technology, Delhi — *New Delhi, India*

- Developed a per-pixel signal processing pipeline in Python (OpenCV, NumPy, SciPy) to detect subsurface defects in Carbon and Glass Fibre-Reinforced Polymer (CFRP/GFRP) composites using active thermography.
- Implemented linear detrending, full temporal cross-correlation against a sound reference pixel, main lobe extraction, and FFT phase computation across three experimental datasets (LFM, Active Pulsed, and Flux excitation).
- Demonstrated that the FFT phase of the correlation main lobe produces qualitatively distinct signatures for defective vs. healthy material regions, establishing a new discriminating observable for non-destructive evaluation. Recorded results and findings in a research paper.

Summer Research Intern | Lunar Rover Modeling — June 2023 – Aug. 2023
NASA California Space Grant Consortium — *Costa Mesa, CA*

- Designed and built a model lunar lava tube exploration rover using a rocker-bogie chassis, solar-powered with lithium-polymer battery storage.
- Developed Arduino firmware integrating UV, Hall effect magnetic, and gravimeter sensors to detect water-ice signatures, iron-rich rock, and subsurface density anomalies.
- Implemented on-board SD card data logging for sensor telemetry and an object avoidance drive system for autonomous navigation over loose regolith.

Projects

Tiled Matrix Multiplication Visualizer | *HTML, JS, Parallel Programming* — Feb. 2026

- Built an interactive visualizer to demonstrate tiled matrix multiplication, data reuse, and memory access patterns.
- Published code on github and deployed at aypuri.com/tiledmatmultvis.

ESP32 Bluetooth Speaker | *Microcontrollers, Circuit Design, C++* — Apr. 2025

- Built a Bluetooth speaker with a display showing title, artist, and album, powered by an ESP32 microcontroller.
- Published code on github

Technical Skills

Languages: C/C++, Python, SystemVerilog, JavaScript, MIPS Assembly
Frameworks & Libraries: OpenCL, PyTorch, OpenCV, CUDA, NumPy/SciPy, scikit-learn
Platforms & Hardware: NVIDIA Jetson, Arduino, ESP32, ARM64 (Pixel Fold)
Tools: Git, Linux, Kubernetes, Docker, MATLAB, Cadence Virtuoso